First Home Assignment
FINA 60201A

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Part 1: Cross-Sectional Analysis

Question 1: Zero-coupon yield extraction

In this question, we build a zero-coupon (spot) curve from the set of money-market instruments and coupon bonds provided in the dataset. We work under the usual no-arbitrage assumption and we follow the convention of *semi-annual* coupon payments for the coupon-bearing bonds. The objective is to recover discount factors $P(0,T)$ on the relevant maturities and then convert them into continuously-compounded zero yields.

We start with Treasury bills, which are already zero-coupon instruments. For a bill maturing at time T with a money-market (simple annualized) yield $y(T)$ (in decimals), the discount factor is computed using simple-interest discounting:

$$P(0,T) = 1 - y(T)\,T. \tag{1}$$

This gives us the short end of the curve directly, without any bootstrapping.

For coupon bonds, we use the fact that a par bond must be priced at 1 (per unit of face value) when discounted using the correct term structure. Consider a bond of maturity T years, paying coupons semi-annually. If the quoted par yield is $y^{\text{par}}(T)$, the semi-annual coupon payment is $c(T) = \frac{y^{\text{par}}(T)}{2}$. Let the coupon dates be $t_i = i/2$ for $i = 1, \ldots, 2T$. The par pricing condition can then be written as

$$1 \;=\; \sum_{i=1}^{2T-1} c(T)\,P(0,t_i) \;+\; \big(1 + c(T)\big)\,P(0,T). \tag{2}$$

When we reach a new maturity T, all earlier discount factors $P(0,t_i)$ are assumed known from previous steps, and (2) lets us solve for the unknown $P(0,T)$:

$$P(0,T) \;=\; \frac{1 - \sum_{i=1}^{2T-1} c(T)\,P(0,t_i)}{1 + c(T)}. \tag{3}$$

In practice, par yields are only observed at a limited set of maturities. Since (2) requires discount factors on the full semi-annual coupon grid, we interpolate the missing par yields across maturity (linearly in maturity) and apply the same bootstrapping logic on that completed grid. This provides a consistent set of discount factors at all coupon dates needed for the cashflows.

Finally, once $P(0,T)$ is available, we convert it into a continuously-compounded zero-coupon yield $z(T)$ using

$$z(T) \;=\; -\frac{1}{T}\ln\big(P(0,T)\big). \tag{4}$$

Table 1 reports the extracted discount factors and the corresponding continuously-compounded zero yields at the maturities in the dataset. These spot rates are then used as the observed inputs for the NSS curve fitting in Question 2.

Table 1: Extracted discount factors and continuously-compounded zero yields.

T (years)	Instrument	$P(0,T)$	$z(T)$ (%)
0.0833	Bill	0.996635	4.0463
0.25	Bill	0.990275	3.9090
0.50	Bill	0.981400	3.7550
1.00	Bill	0.965300	3.5316
2.00	Bond	0.931680	3.5383
3.00	Bond	0.899838	3.5180
5.00	Bond	0.833455	3.6435
7.00	Bond	0.763847	3.8484
10.00	Bond	0.661728	4.1290
20.00	Bond	0.377540	4.8704

Question 2: Nelson–Siegel–Svensson fit of the zero-coupon curve

In this question, we fit a smooth parametric curve to the continuously-compounded zero-coupon yields extracted in Question 1. We use the Nelson–Siegel–Svensson (NSS) specification for spot rates, which is flexible enough to reproduce typical yield curve shapes while keeping a small number of parameters. The main idea is to represent the term structure with level, slope, and curvature components that decay with maturity at speeds controlled by two positive parameters.

Let $z(T)$ denote the observed continuously-compounded zero-coupon yield at maturity T. The NSS model defines the fitted yield $z^{\mathrm{NSS}}(T)$ as

$$z^{\mathrm{NSS}}(T) = \beta_0 + \beta_1 \left(\frac{1 - e^{-T/\tau_1}}{T/\tau_1} \right) + \beta_2 \left(\frac{1 - e^{-T/\tau_1}}{T/\tau_1} - e^{-T/\tau_1} \right) + \beta_3 \left(\frac{1 - e^{-T/\tau_2}}{T/\tau_2} - e^{-T/\tau_2} \right), \tag{5}$$

where $\tau_1 > 0$ and $\tau_2 > 0$. In this representation, β_0 controls the long-run level of rates, β_1 mostly affects the short-end slope, and (β_2, β_3) add curvature contributions at intermediate maturities.

We estimate the parameter vector $\big(\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2\big)$ by minimizing the sum of squared fitting errors on yields over the set of observed maturities $\{T_i\}_{i=1}^n$:

$$\min_{\beta_0,\beta_1,\beta_2,\beta_3,\tau_1,\tau_2} \quad \sum_{i=1}^{n} \big(z(T_i) - z^{\mathrm{NSS}}(T_i) \big)^2 . \tag{6}$$

Since the problem is non-linear in (τ_1, τ_2), we solve it numerically and use multiple starting values to reduce sensitivity to initialization.

Table 2 reports the estimated NSS parameters, together with the minimized objective value. Overall, the fitted curve tracks the observed zero-coupon yields closely across maturities. Figure 1 shows the observed yields and the NSS fitted curve, as well as the residuals $z(T_i) - z^{\mathrm{NSS}}(T_i)$. The residuals remain small and do not display a clear systematic pattern, which indicates that the NSS specification captures the main shape of the extracted zero-coupon curve.

Table 2: NSS parameter estimates and fit objective.

Parameter	Estimate
β_0 (%)	5.913196
β_1 (%)	-1.766070
β_2 (%)	-2.298248
β_3 (%)	-5.625341
τ_1 (years)	0.734075
τ_2 (years)	3.226199
SSE $\sum_{i=1}^{n}\left(z(T_i)-z^{\mathrm{NSS}}(T_i)\right)^2$	0.00401603

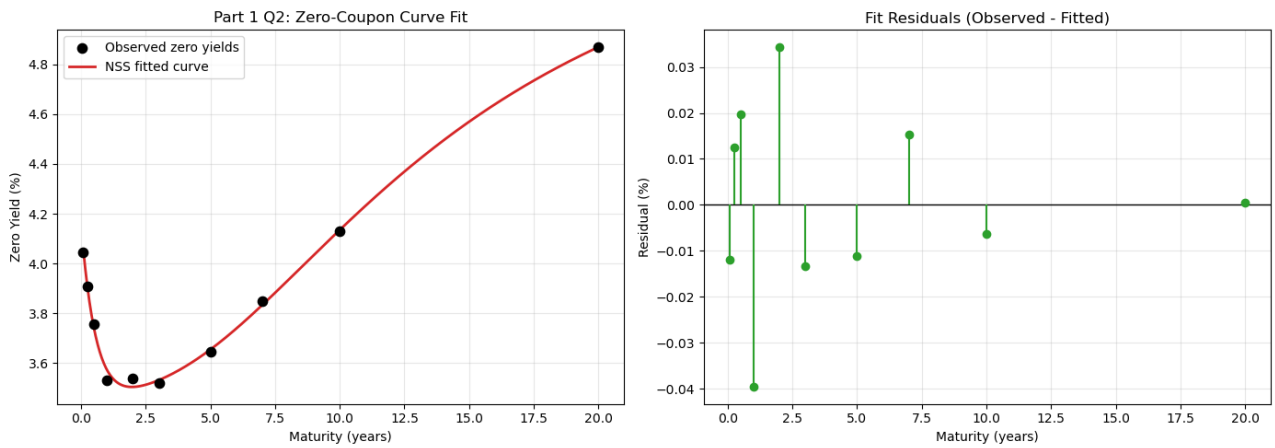

Figure 1: NSS fit of the zero-coupon curve (left) and residuals (right).

Question 3: Calibration of the two-factor Gaussian model to cap prices

In this question, we calibrate a two-factor Gaussian (2FG) short-rate model to the cap market. The market provides cap implied volatilities for maturities $T \in \{1, \ldots, 10\}$ years and for three strikes defined relative to the ATM forward swap rate K_f, namely $K \in \{0.85K_f,\ 1.00K_f,\ 1.15K_f\}$. The calibration is performed on *cap prices*, so we first convert the quoted implied volatilities into market cap prices using the Black framework.

For a given strike K and maturity T, we compute the market cap price $\mathrm{Cap}^{\mathrm{mkt}}(K,T)$ as the sum of Black caplet prices over the reset dates. Under the 2FG model, we price caps using the standard replication of each caplet payoff by an option on a zero-coupon bond. In particular, a cap is valued as a sum of puts on zero-coupon bonds, where the strike of each bond option is determined by the accrual period and the cap strike. Since the 2FG model implies closed-form prices for zero-coupon bond options, the cap price $\mathrm{Cap}^{\mathrm{2FG}}(K,T;\Theta)$ is obtained by summing these bond-put values.

Following the in-class suggestion, we extend the calibration by allowing the current factor states (x_t, y_t) to be estimated jointly with the structural parameters. Hence, we use

$$\Theta = (a, b, \sigma, \eta, \rho, x_t, y_t).$$

For reproducibility, we used a multi-start least-squares calibration with parallel evaluation (10 starting values for the 7-parameter model).

This extension is directly reflected in the bond pricing formula used in class. The model-implied zero-

coupon bond price can be written as

$$P^{2\mathrm{FG}}(t,T) = \frac{P^m(0,T)}{P^m(0,t)} \exp\big(B(t,T)\big), \tag{7}$$

where $P^m(\cdot,\cdot)$ denotes the observable (market) discount factors. The state dependence enters through

$$B(t,T) = \frac{1}{2}\Big(V(t,T) - V(0,T) + V(0,t)\Big) - \underbrace{\frac{1-e^{-a(T-t)}}{a}}_{B_a(t,T)} x_t - \underbrace{\frac{1-e^{-b(T-t)}}{b}}_{B_b(t,T)} y_t. \tag{8}$$

In (8), $V(t,T)$ denotes the 2FG variance term from the lecture notes (the variance of the integrated factor sum over $[t,T]$), and it depends on (a,b,σ,η,ρ).
The parameters are estimated by minimizing the sum of squared *relative* price errors across all instruments, as required:

$$\min_{\Theta} \sum_{j=1}^{3}\sum_{k=1}^{10} \left(\frac{\mathrm{Cap}^{2\mathrm{FG}}(K_j,T_k;\Theta) - \mathrm{Cap}^{\mathrm{mkt}}(K_j,T_k)}{\mathrm{Cap}^{\mathrm{mkt}}(K_j,T_k)} \right)^2. \tag{9}$$

Table 3 reports the calibrated parameters and the minimized objective value. In our run, the best solution is obtained after multiple starting values, and the fitted prices track market prices closely. The extension (x_t, y_t) improves the relative SSE compared to the baseline calibration with $(x_t, y_t) = (0,0)$: the 5-parameter objective is 0.0266612546, while the 7-parameter objective is 0.0146562682. As a diagnostic, we also compute model-implied Black volatilities by inverting the Black formula on the fitted prices. The overall fit is summarized by the cap price RMSE and the implied-vol RMSE reported in Table 4. Figure 2 shows the price fit (which is the calibration target), while Figure 3 reports the implied-vol fit as a secondary check.

Table 3: Calibrated 2FG parameters (including fitted states) and minimized relative-price objective.

Parameter	Estimate
a	0.029483
b	2.758924
σ	0.011792
η	0.028207
ρ	-0.999000
x_t	-0.000297
y_t	0.002860
Minimized objective value in (9)	0.0146562682

Table 4: Fit quality summary (prices and implied volatilities).

Metric	Value
Cap price RMSE	0.00065417
Implied-vol RMSE (percentage points)	0.7624

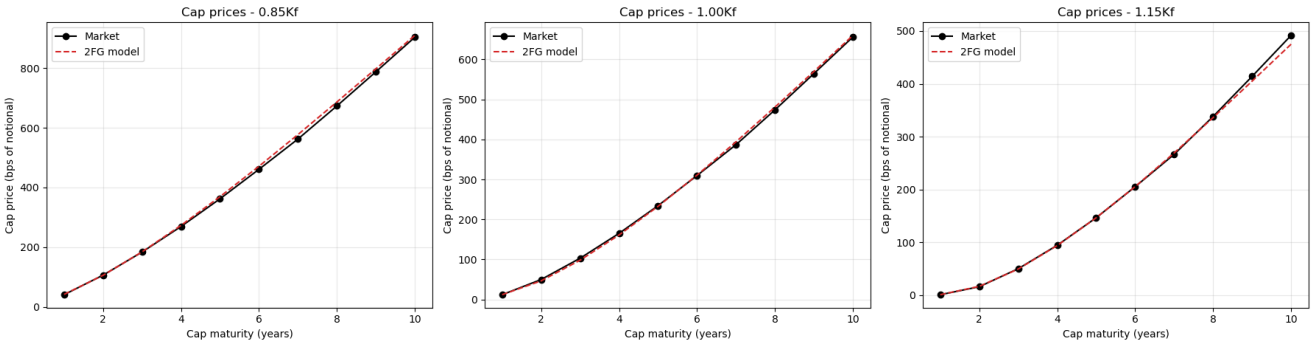

Figure 2: Market versus model cap prices across maturities for strikes $0.85K_f$, $1.00K_f$, and $1.15K_f$.

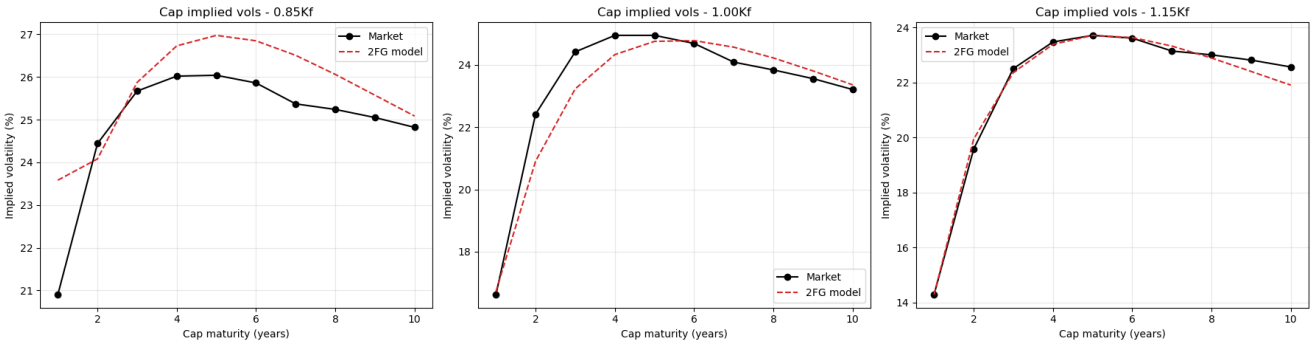

Figure 3: Market versus model cap implied volatilities across maturities for strikes $0.85K_f$, $1.00K_f$, and $1.15K_f$.

Finally, we note that the calibration targets prices through (9), so small differences can remain in implied volatilities even when the price fit is strong. In our calibration, the correlation parameter ρ still reaches the lower bound of the search region, which suggests the optimizer prefers a very negative correlation to match the cross-strike term structure of cap prices. Despite this, the resulting fitted prices remain close to the market across all maturities and strikes considered.

Part 2: Time-Series Analysis

Question 1: Weekly zero-coupon yield series from NSS coefficients

In this question, we reconstruct weekly zero-coupon yields at a fixed set of maturities using the Nelson–Siegel–Svensson (NSS) coefficients provided in the Excel file. For each week t, the dataset contains the parameter vector $(\beta_{0,t}, \beta_{1,t}, \beta_{2,t}, \beta_{3,t}, \tau_{1,t}, \tau_{2,t})$. We treat these as inputs and compute the corresponding continuously-compounded zero-coupon yield $z_t(T)$ for each required maturity T by plugging the weekly coefficients into the NSS yield function:

$$z_t(T) = \beta_{0,t} + \beta_{1,t}\left(\frac{1-e^{-T/\tau_{1,t}}}{T/\tau_{1,t}}\right) + \beta_{2,t}\left(\frac{1-e^{-T/\tau_{1,t}}}{T/\tau_{1,t}} - e^{-T/\tau_{1,t}}\right) + \beta_{3,t}\left(\frac{1-e^{-T/\tau_{2,t}}}{T/\tau_{2,t}} - e^{-T/\tau_{2,t}}\right). \quad (10)$$

We compute $z_t(T)$ for the seven maturities required in the assignment: $T \in \{0.25,\ 0.5,\ 1,\ 3,\ 5,\ 10,\ 30\}$ years, corresponding to 3-month, 6-month, 1-year, 3-year, 5-year, 10-year, and 30-year yields. This produces a weekly panel of yields (in percentage points) over the full sample period in the Excel file.
The sample contains 338 weekly observations, from 2019-08-07 to 2026-01-21. Reported standard deviations are sample standard deviations (ddof= 1).
Finally, for each maturity, we summarize the time-series behavior using descriptive statistics. Table 5 reports the mean, standard deviation, and the first, second, and third quartiles of the weekly yield series for each maturity.

Table 5: Descriptive statistics of weekly NSS-implied zero-coupon yields (in %).

Maturity	Mean	Std Dev	Q1 (25%)	Median (50%)	Q3 (75%)
3-month	2.842952	2.160096	0.218009	3.639531	4.931027
6-month	2.815641	2.090582	0.219845	3.689847	4.789102
1-year	2.752599	1.976020	0.245916	3.579286	4.539148
3-year	2.641048	1.671097	0.733470	3.474668	4.093740
5-year	2.678781	1.513737	1.115698	3.436988	3.993853
10-year	2.925587	1.384548	1.617464	3.435139	4.224936
30-year	3.432659	1.224451	2.237609	3.757467	4.568139

Question 2: Estimation of a one-factor CIR model with the Kalman filter

In this question, we estimate a one-factor CIR short-rate model using weekly zero-coupon yields. The state variable is the (latent) short rate r_t, and the observed data are the yields at the seven maturities $T \in \{0.25, 0.5, 1, 3, 5, 10, 30\}$. We work with a state-space representation where the transition equation comes from the discrete-time approximation of the CIR dynamics and the measurement equation links yields to the short rate through the affine bond pricing formula.
The CIR model assumes that under the risk-neutral measure the short rate follows a square-root diffusion of the form

$$dr_t = \kappa(\theta - r_t)\,dt + \sigma\sqrt{r_t}\,dW_t^{\mathbb{Q}}, \quad (11)$$

with parameters $\kappa > 0$, $\theta > 0$, and $\sigma > 0$. In the estimation, the market price of risk parameter λ enters the state transition under the physical measure through

$$\kappa^{\mathbb{P}} = \kappa - \sigma\lambda, \qquad \theta^{\mathbb{P}} = \frac{\kappa\theta}{\kappa^{\mathbb{P}}},$$

so the Kalman filter dynamics use $(\kappa^{\mathbb{P}}, \theta^{\mathbb{P}}, \sigma)$, while bond pricing and measurement loadings $(a(T), b(T))$ are computed from the risk-neutral parameters (κ, θ, σ). This model implies an exponential-affine expression for zero-coupon bond prices,

$$P(t,T) = A(t,T)\exp\big(-B(t,T)\,r_t\big), \tag{12}$$

where, letting $\Delta = T - t$ and $\gamma = \sqrt{\kappa^2 + 2\sigma^2}$,

$$B(t,T) = \frac{2\left(e^{\gamma\Delta} - 1\right)}{(\gamma+\kappa)\left(e^{\gamma\Delta} - 1\right) + 2\gamma}, \tag{13}$$

$$A(t,T) = \left(\frac{2\gamma\,\exp\big(\frac{\kappa+\gamma}{2}\Delta\big)}{(\gamma+\kappa)\left(e^{\gamma\Delta} - 1\right) + 2\gamma}\right)^{\frac{2\kappa\theta}{\sigma^2}}. \tag{14}$$

This leads to an affine yield representation,

$$y_t(T) = -\frac{1}{T-t}\ln P(t,T) \;=\; a(T) + b(T)\,r_t, \tag{15}$$

with

$$a(T) = -\frac{1}{T-t}\ln A(t,T), \qquad b(T) = \frac{1}{T-t}B(t,T). \tag{16}$$

We then specify the measurement equation for each maturity as

$$y_t(T_i) = a(T_i) + b(T_i)\,r_t + \varepsilon_{t,i}, \qquad \varepsilon_{t,i} \sim \mathcal{N}(0, h_{T_i}), \tag{17}$$

and we estimate the model by maximizing the Gaussian log-likelihood produced by the Kalman filter. Since the state variable is one-dimensional, the filter is straightforward to implement and yields both parameter estimates and filtered values of r_t. (The reported log-likelihood is the Gaussian likelihood from the prediction errors; constants do not affect the maximizer.)

Table 6 reports the estimated parameters, their standard errors, and the maximized log-likelihood value. The estimates are economically reasonable and the standard errors are small for (κ, θ, σ), while the standard error for λ is larger, which is consistent with risk-premium parameters often being harder to pin down precisely from yields alone.

Table 6: CIR parameter estimates (weekly data), standard errors, and maximized log-likelihood.

Parameter	Estimate	Std. Error
κ	0.111707	0.004852
θ	0.032525	0.000744
λ	-0.734511	1.415408
σ	0.050340	0.001934
Maximized log-likelihood	12329.589308	—

As an additional diagnostic, we compute the measurement errors $\varepsilon_{t,i}$ implied by the filter and summarize them by maturity. Table 7 reports the mean and variance of the measurement errors for each maturity (both in decimals and in percentage points), along with the estimated measurement variance h_{T_i}. Overall, the errors remain small, and the variances increase with maturity, which is consistent with longer-term yields being harder to match perfectly with a one-factor model.

Table 7: Measurement error summary by maturity.

Maturity	Mean error (pp)	Var error (pp^2)	h_T (pp^2)	Mean error (decimal)
3-month	0.032344	0.032887	0.034718	0.000323
6-month	-0.000838	0.000058	0.000291	-0.000008
1-year	-0.074905	0.040393	0.045317	-0.000749
3-year	-0.222339	0.207857	0.255281	-0.002223
5-year	-0.210312	0.237563	0.279921	-0.002103
10-year	-0.000864	0.265323	0.263916	-0.000009
30-year	0.471147	0.563632	0.783695	0.004711

Question 3: Filtered CIR short rate versus the NSS short rate

In this question, we compare the short-rate dynamics implied by the one-factor CIR model to the short end implied by the NSS curve. From Question 2, the Kalman filter provides a weekly filtered estimate of the latent short rate $\widehat{r}_t$. To build a comparable NSS-based short rate, we use the fact that the NSS yield function admits a well-defined short-maturity limit. In particular, as $T \downarrow 0$, the NSS yield converges to

$$z_t(0) \equiv \lim_{T \downarrow 0} z_t(T) = \beta_{0,t} + \beta_{1,t}, \tag{18}$$

which we interpret as the NSS-implied short rate. We denote this series by $r_t^{\text{NSS}} = \beta_{0,t} + \beta_{1,t}$.
We then plot $\widehat{r}_t$ and r_t^{NSS} over the full sample, together with the spread $\widehat{r}_t - r_t^{\text{NSS}}$. The two series move together over time and capture similar broad cycles in short-term rates. At the same time, there are periods where the spread widens, reflecting both model differences (one-factor CIR restrictions) and short-end movements implied by the NSS coefficients.

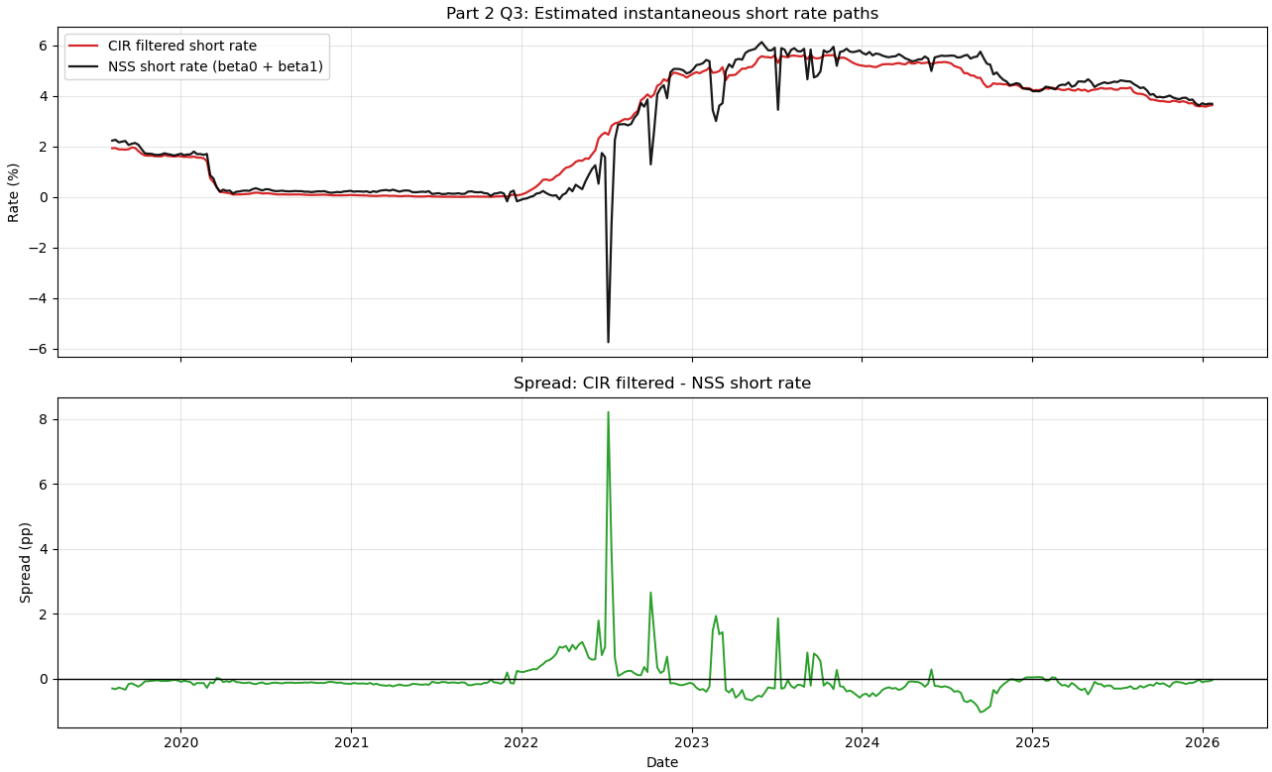

Figure 4: Comparison of the filtered CIR short rate $\widehat{r}_t$ and the NSS-implied short rate $r_t^{\text{NSS}} = \beta_{0,t} + \beta_{1,t}$, together with their spread.

The spread shows a few short-lived spikes, which correspond to abrupt changes in the NSS short-end coefficients (especially $\beta_{1,t}$). The CIR filtered short rate remains smoother in those periods, which highlights the stabilizing role of the filter combining time-series persistence with information across maturities.